

Pool Leak Detection Crash Course

A randomized PDF workbook for interview preparation, accelerated trade literacy, and customer-facing repair vocabulary.

Randomized version

Generated July 01, 2026. Quiz seed: 20260701-3a4d24.

Training note

This is interview preparation and trade-literacy material. It is not a substitute for field training, licensing, electrical safety procedure, diving safety, excavation procedure, or employer SOPs.

What You Should Be Able To Discuss

- How water moves through a pool circulation system.
- How technicians separate evaporation, structural leaks, plumbing leaks, and equipment leaks.
- What bucket testing, dye testing, and pressure testing are used for.
- How concrete/gunite shell repair differs from plaster or pool finish repair.
- How to explain diagnostic findings and repair recommendations in plain language.

Fast Study Map

1. **Pool Systems 101** - Foundation, 8 min
2. **Understanding Water Loss** - Triage, 7 min
3. **The Bucket Test** - First test, 8 min
4. **Leak Detection Logic** - Detective work, 9 min
5. **Dye Testing** - Visual confirmation, 7 min
6. **Pressure Testing** - Plumbing isolation, 9 min
7. **Common Leak Locations** - Pattern recognition, 6 min
8. **Concrete and Guniting Shell Repair Basics** - Repair literacy, 9 min
9. **Pool Finish and Plaster Repair Basics** - Finish repair, 10 min
10. **Customer Psychology** - Trust, 8 min
11. **Service Business Context** - Company fit, 7 min
12. **Interview Vocabulary and Answers** - Performance, 10 min

Six facts to memorize

Pools evaporate. Bucket tests separate evaporation from leaks. Pressure testing checks plumbing. Dye testing confirms visible leak points. Shell and finish defects are different repair conversations. Good technicians think like detectives.

Pool Systems 101

A pool is a contained body of water connected to circulation plumbing. Your first interview win is being able to trace the water path without getting lost.

**Say this
naturally**

I understand the basic circulation path: pool water is pulled through suction points, moved by the pump, cleaned by the filter, optionally heated, and returned through jets.

What To Know

- Skimmers pull surface water and debris into the suction side. Common leak points include cracked skimmer throats and separation from the pool shell.
- Main drains pull from the bottom. They are less common suspects than skimmers or lights, but they matter because access is harder.
- The pump is the heart of circulation. Air leaks on the suction side can show up as bubbles, poor prime, or visible issues at the lid and seals.
- Filters remove suspended debris. Cartridge, sand, and DE filters all serve the same core purpose with different media.
- Return lines push cleaned water back to the pool through fittings or jets. Return fittings and underground return plumbing can leak.

Technician Thinking

- Name the side of the system: suction before the pump, pressure after the pump.
- Separate visible equipment leaks from buried plumbing leaks.
- Do not diagnose from one clue. Use the clue to pick the next test.

Module Quick Check

M1. Which path best describes normal pool circulation?

- A.** Pump to pool to skimmer to drain to heater
- B.** Pool to returns to filter to skimmer to pump
- C.** Pool to skimmer/drain to pump to filter to heater to returns

Your answer: _____

Understanding Water Loss

Not every water drop is a leak. A technician first decides whether the loss is normal evaporation, splash-out, rain overflow, or a true leak.

Say this naturally

Before assuming a leak, I would want to confirm abnormal water loss and account for evaporation, splash-out, wind, rain, and any autofill system.

What To Know

- Evaporation changes with heat, wind, humidity, sun exposure, and water features.
- An autofill can hide a leak because it replaces water while the customer sees a normal level.
- A loss of about one eighth to one quarter inch per day can be normal in many Florida conditions; more than that deserves closer testing.
- A water level that drops to a feature and then slows can point toward that feature: skimmer, light, tile line, or fitting.
- Wet soil, sinking deck areas, air bubbles, or equipment-pad puddles are clues, not final answers.

Technician Thinking

- Ask when the loss happens: pump on, pump off, after rain, after high wind, or constantly.
- Ask what changed recently: repair, resurfacing, new equipment, storm, heavy use.
- Look for the simplest visible issue before assuming underground plumbing.

Module Quick Check

M2. Why can an autofill system make a leak harder to notice?

- A.** It makes dye testing unnecessary
- B.** It permanently seals small plumbing leaks
- C.** It replaces lost water and can hide the true loss rate

Your answer: _____

The Bucket Test

The bucket test compares pool water loss against a small control sample exposed to the same weather. It is one of the simplest ways to separate evaporation from leaking.

Say this naturally

The bucket test helps distinguish evaporation from actual pool loss by comparing the pool level to a bucket of pool water over about 24 hours.

What To Know

- Bring the pool to normal operating level and turn off autofill.
- Put pool water in a bucket and place the bucket on a step so it sits in the same conditions.
- Mark the water level inside the bucket and the pool level outside the bucket.
- Compare after about 24 hours. Rain or a long delay can make the test unreliable.
- If the pool drops more than the bucket, the extra drop is evidence of a likely leak.

Technician Thinking

- The bucket is the control. Both should evaporate similarly.
- The test confirms likely leaking; it does not locate the leak.
- A positive bucket test sends the technician into isolation and location work.

Module Quick Check

M3. In a bucket test, what does it mean if the pool drops more than the bucket?

- A.** Likely leak beyond evaporation
- B.** The filter is definitely broken
- C.** Normal evaporation only

Your answer: _____

Leak Detection Logic

Good technicians do not guess. They run a clean diagnostic sequence: confirm the leak, separate structure from plumbing, narrow the location, verify, then recommend.

**Say this
naturally**

I would think of leak detection as a diagnostic process: confirm the loss, isolate the system, test likely areas, verify the finding, then explain the repair path.

What To Know

- Step 1: Confirm that abnormal loss exists.
- Step 2: Decide whether clues point toward structure, plumbing, equipment, or multiple issues.
- Step 3: Isolate lines, fittings, features, and suspected shell penetrations.
- Step 4: Verify with a second method when possible, especially before excavation.
- Step 5: Explain what was tested, what was ruled out, what remains likely, and what should happen next.

Technician Thinking

- A test that rules something out is still valuable.
- Use the customer's story, but do not let it trap the diagnosis.
- Confidence comes from sequence, not bravado.

Module Quick Check

M4. What is the safest interview framing for leak detection?

- A.** A quick guess based on the first visible crack
- B.** A systematic diagnostic process
- C.** A repair-first trade where testing is optional

Your answer: _____

Dye Testing

Dye testing is useful when you already have a suspected visible leak point. Still water matters because the technician is watching whether dye gets pulled into a gap.

**Say this
naturally**

**Dye testing helps visually confirm suspected leak points around
skimmers, lights, fittings, cracks, and other accessible openings.**

What To Know

- Best use: visible or reachable features such as skimmers, lights, returns, steps, cracks, and fittings.
- Poor use: broad unexplained water loss with no suspected area, moving water, wind, or hidden underground plumbing.
- A positive result looks like dye being pulled into a specific opening rather than drifting randomly.
- Dye can support a repair recommendation, but weak or confusing movement may require another test.
- Pool lights involve electrical safety. An interview answer should show caution and respect for training.

Technician Thinking

- Dye confirms a suspected location; it does not magically search the whole pool.
- Turn circulation off and let water get still before testing when procedure allows.
- Use dye as evidence, not theater.

Module Quick Check

M5. Where is dye testing most useful?

- A.** Only at the equipment pad
- B.** At visible suspected leak points
- C.** Inside every underground pipe

Your answer: _____

Pressure Testing

Pressure testing isolates pool plumbing lines. If a sealed line loses pressure, the technician has evidence that the line may be compromised.

**Say this
naturally**

Pressure testing lets technicians isolate plumbing issues before excavation by sealing a line, applying controlled pressure, and watching whether it holds.

What To Know

- Lines are isolated with plugs so the technician can test one section at a time.
- A gauge that holds steady suggests the line is likely intact under that test condition.
- A pressure drop suggests a possible leak in that isolated line or test setup.
- Results should be interpreted with training because valves, plugs, air, water, and setup details matter.
- Pressure testing is not casual DIY work. It requires proper equipment, procedure, and safety judgment.

Technician Thinking

- Use pressure testing to distinguish plumbing from structure.
- A failed test narrows the search; it does not always pinpoint the exact buried break by itself.
- Verification matters before recommending excavation.

Module Quick Check

M6. What does a pressure drop usually indicate during an isolated line test?

- A.** The isolated line or setup is not holding pressure
- B.** The pool is only evaporating
- C.** The skimmer basket is full

Your answer: _____

Common Leak Locations

You do not need to memorize every rare failure. Start with the common areas and learn why they fail.

**Say this
naturally**

I would pay special attention to skimmers, lights, return fittings, visible cracks, main drains, and the equipment pad because they are common or high-impact leak areas.

What To Know

- Skimmers are common because plastic, concrete, plaster, and movement meet at the throat.
- Pool lights are common because the niche, conduit, and sealing points create paths for water.
- Return fittings can leak around the fitting face or in the buried return line.
- Expansion cracks and shell cracks matter, especially in older pools or after movement.
- The equipment pad can create a false pool leak if water is visibly leaking from pumps, filters, heaters, valves, or unions.

Technician Thinking

- Common does not mean certain.
- Water level clues help prioritize features.
- Check the equipment pad before chasing hidden problems.

Module Quick Check

M7. Which is a common leak area because materials meet and separate?

- A.** Fence gate
- B.** Skimmer throat
- C.** Pool thermometer

Your answer: _____

Concrete and Guniting Shell Repair Basics

Concrete, gunite, and shotcrete pools have a structural shell under the interior finish. Repair thinking starts by deciding whether the issue is structural shell movement, weak material, or only surface finish distress.

**Say this
naturally**

I know a concrete or gunite repair starts with diagnosis: is this a surface finish issue, a bond failure, weak shell material, or a structural crack that needs a trained repair approach?

What To Know

- Gunite and shotcrete are pneumatically placed concrete materials used to form pool shells over steel reinforcement.
- Shell problems can show up as open cracks, recurring leaks, hollow-sounding areas, exposed reinforcement, rust staining, or movement around steps, benches, bond beams, and fittings.
- Sounding is a common concept: solid areas tend to ring differently than hollow, weak, or debonded areas when checked by a trained technician.
- Weak or unsuitable material should not be buried under a cosmetic patch. Sound repair preparation means removing loose or deteriorated material and reaching sound substrate.
- Structural crack repair is not the same as smearing sealant over a crack. It may require engineering judgment, staples, injection systems, waterproofing details, or shell reconstruction depending on severity.

Technician Thinking

- Separate cosmetic symptom from structural cause.
- A leak at a crack should trigger both leak thinking and structural thinking.
- The interview-safe answer is: I would follow company procedure and escalate structural concerns rather than freelancing a repair.

Module Quick Check

M8. What is the key distinction in concrete or gunite repair thinking?

- A. Blue pool versus green pool
- B. Whether the pool has a ladder
- C. Surface finish issue versus shell/structural issue

Your answer: _____

Pool Finish and Plaster Repair Basics

The finish is the visible interior surface: plaster, quartz, pebble, exposed aggregate, or similar cementitious coatings. Finish repair depends on bond, chemistry, prep, cure, color expectations, and whether the damage is local or widespread.

**Say this
naturally**

For pool finish repair, I would think about bond, surface preparation, compatible material, curing/startup, and whether the customer needs a spot repair or a larger resurfacing conversation.

What To Know

- Common finish issues include check cracks, open cracks, etching, scaling, stains, blisters, spalls, delamination, hollow spots, and roughness.
- Delamination means the finish has separated within itself or from the substrate. Small isolated areas may be patched; widespread hollow areas often point toward resurfacing evaluation.
- A durable patch is about preparation, not just material. The repair area should be clean, sound, properly roughened, and compatible with the coating system.
- Color and texture matching are hard. A good customer explanation sets expectations that a patch may stop a defect without becoming invisible.
- Fresh plaster needs careful startup: continuous fill, brushing, balanced water, plaster dust control, and a roughly 28-day early-care window.

Technician Thinking

- Do not promise a perfect color match on aged finish.
- Ask whether the issue is isolated or widespread before recommending patch versus resurfacing.
- Connect repair quality to prep, bond, cure, and water chemistry.

Module Quick Check

M9. Why is a plaster patch not just a cosmetic smear?

- A.** Bond, prep, material compatibility, and curing affect whether it lasts
- B.** Water chemistry does not matter
- C.** Any patch automatically matches forever

Your answer: _____

Customer Psychology

Customers are often frustrated, anxious, and confused. The technician's job is technical and emotional: diagnose, educate, reassure, and recommend.

**Say this
naturally**

**I would want the customer to understand what I tested, what I found,
what it means, and what the next step is, without making them feel
talked down to.**

What To Know

- Customers may already have spent money or heard conflicting opinions.
- A calm explanation is part of the service, not a bonus.
- Specifics build trust: name the test, the result, and the next step.
- Avoid vague conclusions like, 'You need repairs.' Explain the evidence.
- Good communication can turn a stressful house call into a confident repair decision.

Technician Thinking

- Start with what the customer cares about: stop water loss and avoid surprises.
- Translate test results into plain English.
- Use certainty carefully. Say what is confirmed and what is likely.

Module Quick Check

M10. Which customer explanation is stronger?

- A.** The pool is bad.
- B.** You need repairs.
- C.** I pressure tested the return line, saw pressure loss, and the next step is locating and repairing that line.

Your answer: _____

Service Business Context

A leak detection visit is usually part diagnosis, part customer education, and part repair recommendation. In an interview, connect your answers to that full customer journey without relying on any employer-specific language.

**Say this
naturally**

What interests me is that the role combines field diagnosis, customer education, and a practical repair path, not just finding a problem and leaving the customer confused.

What To Know

- A strong service call moves from concern to evidence to next step.
- Customers need simple explanations, not a technical fog bank.
- Repair recommendations should be tied to what was tested and observed.
- A good applicant sounds trainable, precise, physically ready, and comfortable with customers.
- The safest company-fit answer focuses on learning the employer's actual process during training.

Technician Thinking

- Do not over-flatter. Tie the work itself to your working style.
- Ask about training, route structure, KPIs, documentation, and how repairs are handed off.
- Show you understand the business outcome: diagnosis that leads to a confident next step.

Module Quick Check

M11. What should you connect the role to in an interview?

- A.** Diagnosis, customer education, repair options, and follow-through
- B.** Only swimming ability
- C.** Only sales pressure

Your answer: _____

Interview Vocabulary and Answers

The final step is sounding natural. Use trade terms accurately, then connect them to how you think and work.

**Say this
naturally**

Leak detection appeals to me because it is hands-on diagnostic work with a customer-service component: find the root cause, explain it clearly, and help the customer move toward a solution.

What To Know

- Core terms: leak detection, pressure testing, dye testing, plumbing isolation, return line, suction line, skimmer, main drain, equipment pad, structural leak, evaporation loss, gunite, delamination, spall, and startup.
- Use vocabulary only when it helps. Forced jargon sounds worse than plain English.
- Prepare one answer for interest, one for customer conflict, one for learning a new trade, and one for physical field work.
- Ask smart questions: What does training look like? What makes a new technician successful? How are findings documented? What KPIs matter?
- Close by sounding ready to learn: systematic, customer-aware, and coachable.

Technician Thinking

- Memorize phrases, not speeches.
- Pause before technical answers so you sound thoughtful.
- Connect your past experience to the diagnostic and customer parts of the job.

Module Quick Check

M12. Which answer best frames interest in the position?

- A.** I heard pools are easy.
- B.** I do not like talking to customers.
- C.** It combines problem-solving, field work, and customer communication.

Your answer: _____

Vocabulary Glossary

Term	Plain-English meaning
Pressure testing	Sealing and pressurizing an isolated plumbing line to see whether it holds pressure.
Dye testing	Using leak-detection dye near a suspected visible leak point to watch whether water pulls dye into an opening.
Suction line	Plumbing before the pump that pulls water from skimmers or main drains.
Return line	Plumbing after the filter/heater that sends water back into the pool through returns.
Structural leak	A leak in the pool shell, skimmer area, light niche, fitting, crack, or other pool structure.
Equipment pad	The area where pump, filter, heater, valves, unions, and related equipment are located.
Evaporation loss	Water lost naturally to weather conditions rather than through a defect.
Plumbing isolation	Separating one line or system section so it can be tested without confusing results.
Gunite	A pneumatically applied concrete material commonly used to form reinforced pool shells.
Delamination	A bond or layer failure where finish material separates from itself or the substrate.
Spall	A lifted or broken-away upper finish layer, often related to bond, impact, drying, or chemistry issues.
Startup	The early care period after plastering when fill, brushing, water balance, and dust control protect the finish.

Final Readiness Quiz

The choices below are randomized for this PDF version. Aim for 10 or better before the interview. Use the answer key only after a full attempt.

Q1. Which path best describes normal pool circulation?

- A. Pool to skimmer/drain to pump to filter to heater to returns
- B. Pump to pool to skimmer to drain to heater
- C. Pool to returns to filter to skimmer to pump

Your answer: _____

Q2. Why can an autofill system make a leak harder to notice?

- A. It replaces lost water and can hide the true loss rate
- B. It makes dye testing unnecessary
- C. It permanently seals small plumbing leaks

Your answer: _____

Q3. In a bucket test, what does it mean if the pool drops more than the bucket?

- A. Normal evaporation only
- B. Likely leak beyond evaporation
- C. The filter is definitely broken

Your answer: _____

Q4. What is the safest interview framing for leak detection?

- A. A quick guess based on the first visible crack
- B. A systematic diagnostic process
- C. A repair-first trade where testing is optional

Your answer: _____

Q5. Where is dye testing most useful?

- A. Only at the equipment pad
- B. Inside every underground pipe
- C. At visible suspected leak points

Your answer: _____

Q6. What does a pressure drop usually indicate during an isolated line test?

- A. The isolated line or setup is not holding pressure
- B. The pool is only evaporating
- C. The skimmer basket is full

Your answer: _____

Q7. Which is a common leak area because materials meet and separate?

- A. Pool thermometer
- B. Fence gate
- C. Skimmer throat

Your answer: _____

Q8. What is the key distinction in concrete or gunite repair thinking?

- A. Surface finish issue versus shell/structural issue
- B. Whether the pool has a ladder
- C. Blue pool versus green pool

Your answer: _____

Q9. Why is a plaster patch not just a cosmetic smear?

- A. Bond, prep, material compatibility, and curing affect whether it lasts
- B. Any patch automatically matches forever
- C. Water chemistry does not matter

Your answer: _____

Q10. Which customer explanation is stronger?

- A. You need repairs.
- B. The pool is bad.
- C. I pressure tested the return line, saw pressure loss, and the next step is locating and repairing that line.

Your answer: _____

Q11. What should you connect the role to in an interview?

- A. Only swimming ability
- B. Only sales pressure
- C. Diagnosis, customer education, repair options, and follow-through

Your answer: _____

Q12. Which answer best frames interest in the position?

- A.** I do not like talking to customers.
- B.** I heard pools are easy.
- C.** It combines problem-solving, field work, and customer communication.

Your answer: _____

Answer Key

Module Quick Checks

M1. C - Pool to skimmer/drain to pump to filter to heater to returns

Water is pulled from the pool by skimmers/main drains, moved by the pump, filtered, optionally heated, and returned.

M2. C - It replaces lost water and can hide the true loss rate

Autofill keeps the level up, so the customer may not see how much water is being lost.

M3. A - Likely leak beyond evaporation

The bucket accounts for evaporation. Extra pool drop points to water leaving somewhere else.

M4. B - A systematic diagnostic process

The strongest signal is that you understand methodical diagnosis.

M5. B - At visible suspected leak points

Dye is visual. It works best where a technician can place dye near a suspected opening.

M6. A - The isolated line or setup is not holding pressure

A dropping gauge means pressure is escaping somewhere in that isolated test path or setup.

M7. B - Skimmer throat

Skimmer throats are classic suspects because the skimmer and shell can separate.

M8. C - Surface finish issue versus shell/structural issue

The repair path depends on whether the defect is in the finish layer, the bond, or the structural shell.

M9. A - Bond, prep, material compatibility, and curing affect whether it lasts

Finish repairs depend on preparation, bond, compatible materials, curing/startup, and later water balance.

M10. C - I pressure tested the return line, saw pressure loss, and the next step is locating and repairing that line.

The strongest explanation names the test, result, and next step.

M11. A - Diagnosis, customer education, repair options, and follow-through

The role sits at the intersection of technical diagnosis and customer trust.

M12. C - It combines problem-solving, field work, and customer communication.

That answer maps to the actual role: diagnostics plus people.

Final Quiz

Q1. A - Pool to skimmer/drain to pump to filter to heater to returns

Water is pulled from the pool by skimmers/main drains, moved by the pump, filtered, optionally heated, and returned.

Q2. A - It replaces lost water and can hide the true loss rate

Autofill keeps the level up, so the customer may not see how much water is being lost.

Q3. B - Likely leak beyond evaporation

The bucket accounts for evaporation. Extra pool drop points to water leaving somewhere else.

Q4. B - A systematic diagnostic process

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Dye is visual. It works best where a technician can place dye near a suspected opening.

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Finish repairs depend on preparation, bond, compatible materials, curing/startup, and later water balance.

Q10. C - I pressure tested the return line, saw pressure loss, and the next step is locating and repairing that line.

The strongest explanation names the test, result, and next step.

Q11. C - Diagnosis, customer education, repair options, and follow-through

The role sits at the intersection of technical diagnosis and customer trust.

Q12. C - It combines problem-solving, field work, and customer communication.

That answer maps to the actual role: diagnostics plus people.

Interview Answer Bank

Technical problem-solving

What interests me is the diagnostic side of the work. A pool leak is not always obvious, so the technician has to confirm the problem, isolate whether it is structural or plumbing, test likely areas, and verify the finding before recommending a repair.

Customer communication

I like that this role is not just technical. The customer is usually frustrated or confused, so a good technician has to explain what was tested, what the result means, and what the next step is in plain language.

Operations background

My operations background helps because leak detection seems to reward process. You follow a sequence, document findings, communicate clearly, and move the customer toward a practical repair decision.

Trainability

I am not walking in pretending to be a leak expert. What I can bring is curiosity, reliability, customer communication, and a willingness to learn the diagnostic process the right way.

References

American Leak Detection: Bucket Test

<https://www.americanleakdetection.com/diy-leak-tests/bucket-test/>

Bucket test procedure and interpretation.

Anderson Manufacturing: Pressure Test Plumbing

<https://www.leaktools.com/pressure-test-plumbing.html>

Pressure testing as a process-of-elimination method for structure versus plumbing leaks.

National Plasterers Council: Pool Plaster FAQ

<https://www.nationalplastererscouncil.com/faq/>

Generic plaster and finish concepts including cracking, bonding, delamination, spalls, and repair judgment.

Pool Engineering: Minor Gunite or Shotcrete Repair

<https://www.pooleng.com/procedures-for-minor-gunite-or-shotcrete-repair-in-swimming-pools-2/>

Generic shell repair concepts including sounding, removing weak material, roughening, cleaning, and bonding.

NPT: Pool Finish Startup Procedure

<https://www.nptpool.com/pool-finishes/start-up-procedure/>

Generic startup concepts after a new plaster finish.